Debopam Goswami

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Research Summary

My research in sub-GeV dark matter (DM) and neutrino phenomenology spans both lab-based and astrophysical searches. I have developed analytical and computational tools for multi-body scattering that have improved sensitivity across a range of neutrino and DM experiments for several dark-sector models. In my ongoing work, I am applying machine learning techniques to enhance signal discrimination. More broadly, I am interested in leveraging artificial intelligence to automate modeling, simulation, and scientific discovery within particle physics and related disciplines.

Education

Texas A&M University (TAMU), College Station — Texas, USA
Doctor of Philosophy (Ph.D.) in Physics — Aug 2021 – Present
Advisor: Prof. Bhaskar Dutta | GPA: 4.0/4.0

Indian Institute of Technology (IIT) Kanpur — Uttar Pradesh, India
Master of Science (M.Sc.) in Physics — July 2019 – May 2021
Thesis: Dark Matter Genesis in the Early Universe
Advisor: Prof. Debtosh Chowdhury | CGPA: 9.3/10.0 | **Rank: 1**

University of Calcutta (CU) — West Bengal, India
Bachelor of Science (B.Sc.) in Physics — July 2016 – May 2019
Aggregate: 78.5% | First Class with Distinction

Publications

1. B. Dutta, **D. Goswami**, B. Hu, W. C. Huang, V. Pandey, *Dark Matter-Induced Nuclear De-Excitation at SBND with Ab Initio Nuclear Theory*, (Submitted to PRL) [arXiv:2602.02817]
2. B. Dutta, **D. Goswami**, A. Karthikeyan, *Monophotons from Scalar Portal Dark Matter at Neutrino Experiments*, J. High Energ. Phys. 2026, 142 (2026) [arXiv:2507.07936]
3. B. Dutta, **D. Goswami**, A. Karthikeyan, K. J. Kelly, *Dirt/Detector/Dump: Complementary BSM production at Short-Baseline Neutrino Facilities*, J. High Energ. Phys. 2025, 240 (2025) [arXiv:2501.09840]
4. S. Akhtar, S. Choudhury, S. Chowdhury, **D. Goswami**, S. Panda, A. Swain, *Open Quantum Entanglement: A study of two atomic system in static patch of de Sitter space*, Eur. Phys. J. C 80, 748 (2020) [arXiv:1908.09929]

<u>SOON TO APPEAR...</u>

5. B. Dutta, **D. Goswami**, A. Karthikeyan, V. Pandey, Z. Tabrizi, A. Thompson, R. G. Van de Water, *Prying Open the Dark Sector Window with SBND Off-Target Mode*, [arXiv:2602.*****]
6. C. Cappiello, P.S.B. Dev, J.B. Dent, B. Dutta, **D. Goswami**, A. Karthikeyan, *Boosted dark matter from single, stacked blazars and cosmic protons*, [arXiv:2602.*****]
7. B. Dutta, **D. Goswami**, J. Kumar, M. Rai, D. Sathyan, *Explanation of Galactic Center GeV Excess via Cosmic Ray-Dark Matter Scattering*, [arXiv:2602.*****]

Research Experiences

1. **Graduate Researcher** Aug 2021 – Present
   Advisor: Prof. Bhaskar Dutta | Mitchell Institute for Fundamental Physics & Astronomy Texas, USA

   Accelerator-Based Dark Matter & HNL Searches

   - Investigated sub-GeV heavy neutral leptons (HNLs) with transition magnetic moments in the Fermilab Short-Baseline Neutrino (SBN) program, emphasizing beam-dump production and exploiting high-Z enhancement in iron dumps.
   - Identified distinctive kinematic signatures from production in dump, dirt, and detector regions to improve signal-to-background discrimination.
   - Improved SBND sensitivity to light dark matter via localized low-energy electromagnetic activity ("blips") using state-of-the-art ab initio nuclear structure calculations including argon excited states up to 18 MeV.
   - Analyzed monophoton signatures from $2 \to 3$ dark matter-nucleus scattering processes, mediated by scalar particle, in DUNE ND, SBN program, and CCM200, incorporating detailed energy, angular, timing, and spatial distributions.

   Astrophysical & Boosted Dark Matter Phenomenology

   - Proposed a novel mechanism for the explanation of Galactic Center gamma-ray excess using cosmic-ray proton scattering off inelastic dark matter or elastic dark matter with 2 mediators in the Milky Way halo.
   - Demonstrated excellent fit with observed Galactic Center gamma-ray excess within allowed parameter space.
   - Derived improved constraints on dark photon and light scalar mediator models using blazar-boosted dark matter scenarios.
   - Performed comprehensive sensitivity analyses across 10 experiments (DUNE FD, IceCube, Hyper-K, JUNO, LZ, PandaX-4T, Borexino, etc.), including both single and stacked blazar analyses.

   Theoretical & Computational Developments

   - Developed scalable Python packages implementing analytical expressions for multi-body ($2 \to 3$ and $2 \to 4$) scattering processes relevant in neutrino, dark matter experiments, and blazar/cosmic boosted dark matter.
   - Developed Python-based computational frameworks for different kinematic spectra modeling, and sensitivity projections across multiple detectors over various BSM models.
   - Automated BIGSTICK implementation for computing nuclear transition strengths across isotopes.

   Ongoing & Future Work

   - Applied machine learning techniques to the SBN program at Fermilab and various astrophysical sources.

2. **Dark Matter Genesis in the Early Universe.** Jan 2020 - May 2021
   Advisor: Prof. Debtosh Chowdhury | Indian Institute of Technology (IIT) Kanpur Uttar Pradesh, India

   - Investigated the mechanisms responsible for non-thermal dark matter production in the early universe.
   - Explored axion as a viable dark matter candidate, analyzing constraints on its mass, coupling strength, and natural emergence in theories beyond the Standard Model.
   - Performed quantitative calculations on relic photon and neutrino abundances, WIMP miracle, dark matter density via the Lee-Weinberg bound, and solutions of the Boltzmann equation.

3. **Perturbative & Non-Perturbative Physics and Instantons.** Jun 2019 - July 2019
   Advisor: Prof. Sayantan Sharma | The Institute of Mathematical Sciences (IMSc) Tamil Nadu, India

   - Calculated vacuum functionals for harmonic and anharmonic oscillators using semi-classical path integral methods. Investigated asymptotic expansions of integrals, asymptotic solutions to differential equations, summation of divergent series, and instanton concepts in quantum mechanics.

- Determined ground state corrections for the anharmonic quartic oscillator using perturbation theory and applied asymptotic relations to the double well potential. Used Borel re-summation technique to extract meaningful results from non-convergent energy perturbation series.

4. **Path Integral Technique in Quantum Field Theory.** Nov 2019 - Dec 2019
   Advisor: Prof. Ashoke Sen | Harish-Chandra Research Institute (HRI) Uttar Pradesh, India
   - Developed expertise in the Feynman Path Integral technique in quantum field theory and studied Feynman diagrams. Applied these methods to Klein-Gordon, interacting scalar, complex scalar, and Dirac field theories.
   - Worked on the fundamentals of quantum electrodynamics and calculated cross sections for various scattering processes.

5. **Linear Algebra and Group Theory.** Sep 2017 - Nov 2017
   Advisor: Prof. Sourov Roy | Indian Association for the Cultivation of Science (IACS) West Bengal, India
   - Developed expertise in group theory with a focus on the properties of S3, normal subgroups, quotient groups, homomorphisms, isomorphisms, automorphisms, conjugacy classes, and permutation groups.

Talks and Presentations

1. **PROSPECT Experiment** : (Online) Jul 2025
   Talk: Probes of BSM Models in PROSPECT Experiment.
2. **Phenomenology Symposium** : University of Pittsburgh May 2025
   Talk: From Neutron Stars to Beam Dumps: 2-to-3 Process with Single Photon Final State. [Link]
3. **Particle Physics on the Plains** : University of Kansas Nov 2024
   Talk: Signals from Cosmic Boosted Strongly Interacting Dark Matter. [Link]
4. **APS DPF-Phenomenology Symposium** : University of Pittsburgh May 2024
   Talk: Utilizing the Iron Dump at SBN Facilities to Probe Heavy Neutral Lepton and Dark Matter. [Link]
5. **Seminar Speaker** : IIT Kanpur April 2021
   Talk: Dark Matter Genesis in the Early Universe.

Conferences and Workshops

(Excluding the conferences where I have delivered a talk or presentation).

1. **The Mitchell Conference on Collider, Dark Matter, & Neutrino Physics.** May 2024
   Texas A&M University
2. **The Mitchell Conference on Collider, Dark Matter, & Neutrino Physics.** May 2023
   Texas A&M University
3. **Physics of the Early Universe (Online)** Aug 2020
   The International Centre for Theoretical Sciences
4. **XVIIITH Annual Basic Sciences Summer School.** May 2016
   Indian Association for the Cultivation of Science (IACS)

Languages

- Bengali - Native
- English - Professional Fluent; Primary medium of instruction.
- Hindi - Advanced; Fluent and comfortable.

Standardized Test Result

- TOEFL - Total: 107/120 Reading: 27, Listening: 27, Speaking: 27, Writing: 26

Technical Skills

- **Programming Languages** – Python, C++, Bash/Shell, SQL, Wolfram Mathematica, R.
- **Libraries** – NumPy, Pandas, SciPy, JAX, Matplotlib, Seaborn, TensorFlow, PyTorch, LangChain, XGBoost, Scikit-learn, Multiprocessing, Git/GitHub.
- **Machine Learning** – Gradient Boosted Trees, Logistic Regression, Random Forest, Multi-Layer Perceptron, Large Language Models, Retrieval-Augmented Generation.
- **Tools** – BIGSTICK, MadGraph, LaTeX, High Performance Computing, Monte Carlo, Jupyter, Microsoft Office.
- **Operating Systems**: macOS, Linux (Ubuntu), Windows.
- **Soft Skills** – Problem solving, critical and independent thinking, attention to detail, teamwork.

Additional Projects

- **Large Language Model (LLM) Powered Scientific Paper Summarizer** Nov 2025 - Present
  Built a Retrieval-Augmented Generation (RAG) system using Meta's FAISS and Gemini LLM to summarize 2.4M+ arXiv scientific papers with sub-second query response times, and implemented a Streamlit UI for interactive querying (Work in Progress...).

Awards and Honors

1. Awarded the **Graduate Student Travel Grant twice** (2024) by Texas A&M University, for participation in conferences, recognizing exceptional academic achievement and research contributions. [Link]
2. Awarded the **General Proficiency Medal 2021** for achieving the **best academic performance** among graduating M.Sc. students in the Department of Physics at IIT Kanpur, India. [Link]
3. Received the **Academic Excellence Award** for two consecutive academic years (2019–2020 and 2020–2021) at IIT Kanpur, acknowledging consistent academic distinction and top-tier performance. [Link]
4. Secured **All India Rank 10** (99.9th percentile) among 14,000+ candidates in Joint Entrance Screening Test (JEST) 2019, the National screening test for Ph.D. programs. [Link]
5. Secured **All India Rank 59** (99.6th percentile) among 16,000+ candidates in IIT-Joint Admission Test (IIT-JAM) 2019, the National screening test for M.Sc. programs in IITs. [Link]
6. Awarded the **Prof. Anil Sen Memorial Prize** by University of Calcutta for achieving the highest score in Physics Practical Examination in B.Sc. 1st, 2nd & 3rd-year exams taken together. [Link]
7. Ranked in **All India Top 20** (99.8th percentile) among 13,000+ candidates in the National Graduate Physics Exam (NGPE), conducted by Indian Association of Physics Teachers (IAPT). [Link]

Teaching Experience

- PHYS 206 University Physics: Newtonian Mechanics for Engineering and Science — Fall 2024, Spring 2022
- PHYS 206 Don't Panic: Newtonian Mechanics for Engineering and Science — Spring 2024
- PHYS 227 Electricity and Magnetism Laboratory for the Science — Fall 2023, Spring 2023
- PHYS 226 Physics of Forces and Motion Laboratory — Spring 2023, Fall 2022
- PHYS 207 Don't Panic: Electricity and Magnetism for Engineering and Science — Summer 2022
- PHYS 207 University Physics: Electricity and Magnetism for Engineers and Scientists — Fall 2021

Certifications

- **Supervised Machine Learning: Regression and Classification -** DeepLearning.AI : [Certificate] — Jul 2023
- **Python Data Structures -** University of Michigan : [Certificate] — Dec 2023

Leadership & Engagement

1. **Texas A&M Graduate Consulting Club (TAMUGC)** [Link] — College Station, Texas
   Pro Bono Project: AI in Action: Redefining Talent Management for a New Era of HR — Feb 2025 - June 2025
   - Led a pro bono consulting project focused on AI-enabled hiring systems, authored a research article, developed a 30-item best practices toolkit, and designed a two-day client workshop.
   - Presented deliverables to senior stakeholders, and refined content based on feedback.
2. **Texas A&M University (TAMU)** [Link] — College Station, Texas
   Graduate Teaching Assistant — Aug 2021 - Dec 2024
   - Supervised 9 courses for over 3 years, mentoring and grading 500+ undergraduate students.
   - Developed strong communication skills by teaching complex physics concepts to diverse audiences with instructional time of 300+ hours.
3. **Indian Graduate Student Association (IGSA)** [Link] — College Station, Texas
   Director, Editorial Team — July 2024 - Oct 2024
   - Managed social media platforms in collaboration with the editorial team to enhance outreach and engagement.
   - Contributed to the design and development of planners for events, ensuring effective organization and execution.

Interests

Outside of research, I am an active member of the Texas A&M Badminton Club. I also enjoy chess, cricket, hiking, photography, and sketching. I maintain a separate gallery showcasing selected hiking photos, photography, and sketches.

References

1. **Prof. Bhaskar Dutta**
   Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, USA
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   Bhaskar Dutta Website

2. **Prof. Bhupal Dev**
   Department of Physics, Washington University in St. Louis, USA
   bdev@wustl.edu
   Bhupal Dev Website

3. **Prof. Kevin Kelly**
   Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, USA
   kjkelly@tamu.edu
   Kevin Kelly Website

4. **Dr. Vishvas Pandey**
   Fermi National Accelerator Laboratory
   vpandey@fnal.gov
   Vishvas Pandey Website